

Monsij Biswal

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SUMMARY OF QUALIFICATIONS

- Highly motivated PhD student in Electrical and Computer Engineering with extensive experience in signal compression and video coding for immersive media.
- Strong technical writing and communication skills with 3 publications.
- Collaborative and team-oriented demonstrated by managing student organizations and leading research projects.

TECHNICAL SKILLS

- **Programming & Frameworks:** C++, C, Python
- **Software:** MATLAB, Inkscape

EDUCATION

MS/PhD, Electrical and Computer Engineering, University of California, Santa Barbara 2020-Present

- *Advisor:* Prof. Kenneth Rose
- *Major:* Communications *Minor:* Machine Intelligence *Award:* Departmental Funding GPA: 3.9/4
- *Relevant Courses:* Image Processing, Signal Compression, Optimal Estimation and Filtering

B.Tech, Electronics and Communication Engineering, National Institute of Technology 2016-2020

- *Awards:* DAAD WISE Scholarship, MITACS Globalink Research Intern Award GPA: 9.55/10
- *Relevant Courses:* Soft Computing, Antenna and Propagation, Software Engineering

RESEARCH EXPERIENCE

Graduate Student Researcher, Signal Compression Lab, UCSB Present

- Working towards compressing VR180 stereoscopic videos with geodesic motion compensation.
- Experience in video coding using HEVC with 360Lib.

Summer Research Intern, Scientific Computing Group, UCSB June 2022 - August 2022

Project: On the exploration of training networks on tiles of satellite images

- Leveraged the performance of tile-based training to achieve near state-of-the-art results for semantic image segmentation.
- Investigated the effects of varying tile size, data augmentation and network architecture on training complexity and overall classification score.

Research Intern, Christian Albrechts Universität zu Kiel, Germany May 2019 - July 2019

Project: Custom Neural Network Training Function for Optical Transmission System

- Analyzed various neural network training algorithms for equalizing a higher order modulation format in an optical transmission system.
- Implemented custom training function and presented research findings to the group.

Undergraduate Researcher, National Institute of Technology Durgapur August 2018 - March 2020

Project: Millimeter Wave Channel Modelling and Performance Indicators

- Channel modelling based on channel sounding measurements in a vehicular & outdoor scenario centered around a specific license free band of 60GHz.
- Extensive analysis of small scale and large scale fading parameters and their effects on rms delay spread of the channel.

LEADERSHIP EXPERIENCE

Vice-President [CSP], ECE Graduate Student Association, UCSB September 2021 - Present

- Lead a team of 4 graduate students by coordinating logistical tasks and delegating responsibilities.
- Facilitated group discussion and meetings to plan research seminars, lightning talks and social hours.